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What If the OWC Was Misplaced? Investigating the Mystery of OWC in the Intracarbonate Reservoir from South Oman Salt Basin

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Abstract

Stringer reservoirs in the south of Sultanate of Oman are isolated carbonate bodies encased within thick Ara salt formations. These unique geological settings present a range of pressure regimes from hydrostatic to lithostatic which pose considerable challenges in reservoir characterization and development. One of the most critical technical uncertainties in these reservoirs is the accurate determination of the oil-water contact (OWC), which directly impacts reserves estimation and development planning.

In this study, the OWC was primarily interpreted using resistivity log responses across multiple wells. A notable anomaly was observed: one well indicated a shallower OWC compared to the oil-down-to (ODT) depth recorded in a nearby offset well. To investigate this inconsistency, a multi-scenario modelling approach was adopted. The study integrated geological, petrophysical, and fluid property data, utilizing structural interpretation and reservoir modelling tools to examine various scenarios, including potential compartmentalization and fluid distribution heterogeneity.

Perched oil-water contacts (OWCs) are a common phenomenon in faulted fold structures, particularly those characterized by asymmetrical limbs with varying lengths and dips. During hydrocarbon migration, water is displaced on both sides of the fault. However, when a fault is present on the shorter limb of the fold, it can act as a barrier, trapping water on that side. Meanwhile, hydrocarbons continue to migrate and displace water on the opposite fold limb, resulting in the development of two distinct OWC levels. This study presents a novel interpretation of perched water in intrasalt stringer reservoirs and emphasizes the critical role of compartmentalization in influencing fluid distribution. Recognizing and correctly identifying perched water can significantly improve the accuracy of reserve estimates and inform more effective development strategies. By demonstrating the value of integrated, multidisciplinary analysis, this work contributes to reduce subsurface uncertainty and enhancing reservoir management in complex carbonate- salt systems.

Introduction

Petroleum Development Oman operates carbonate fields in the southern part of The Sultanate of Oman that are producing sour hydrocarbons. The targeted reservoirs are carbonate stringers positioned in the salt basin of southern Oman ([Fig.1](#)). The Ara group (late pre-Cambrian - early Cambrian age) contains mainly salt,

and the carbonate stringers are "floating" in the salt. This setup keeps the carbonate stringers unattached to any external aquifers, resulting in most of these floaters being over-pressured and self-charging reservoirs. Pressure regimes can range from hydrostatic to near-lithostatic, adding further complexity to OWC identification and reservoir modelling (Petroleum Development Oman, 2025). Accurate identification of the oil-water contact (OWC) is critical for robust reserves estimation, well placement, and long-term reservoir management. In geologically complex and pressure-isolated carbonate systems, such as those found in the South Oman Salt Basin, OWC determination remains one of the most significant technical uncertainties.

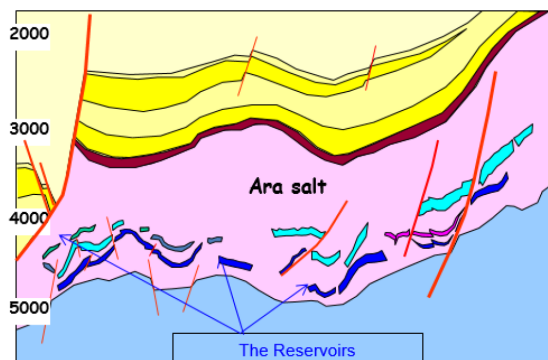


Figure 1—Schematic cross-section of the Ara Group salt basin in South Oman, showing carbonate stringer reservoirs encased within the Ara. (Petroleum Development Oman, 2025)

This study aims to explain the observed OWC variation in the Z field by integrating geological, structural, and pressure data. Focus is given to understanding compartmentalization effects, potential stringer separation, and their implications on fluid distribution.

Z Field Overview

Z field contains hydrocarbons in a carbonate reservoir which is part of Ara group (Late pre-Cambrian - Early Cambrian age). The Ara formation is comprised of Salt and Carbonate bodies 'floating' in the salt deposited during a period of rapid subsidence in the salt basin. The carbonate reservoirs ('stringers') were deposited during Transgressive Systems Tracts (TST) and High Stand Systems Tracts (HST). Within the Ara group, there are six cycles of carbonate bodies (Fig.2). Z field is classified as an A2 stringer consists of mainly limestone and dolomite lithofacies (Petroleum Development Oman, 2025).

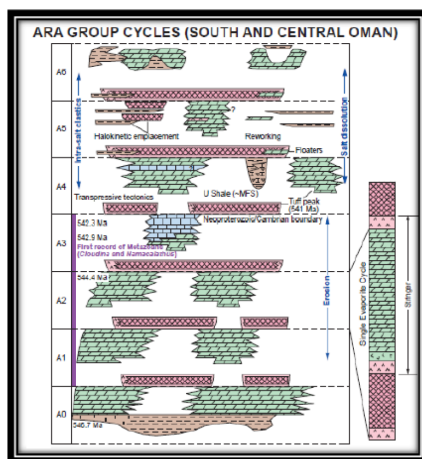


Figure 2—Ara Group stratigraphy showing six carbonate cycles (A1-A6) (Petroleum Development Oman, 2025).

The field was discovered by Z-1H3 in 2001 and has been appraised & developed by 25 wells (Fig.3). There initial oil gradient was acquired in the well Z-1H3 (Spud date 9-Jan-2001), but unfortunately there is no water gradient acquired in the field to derive Free Water Level (FWL) based on pressure data.

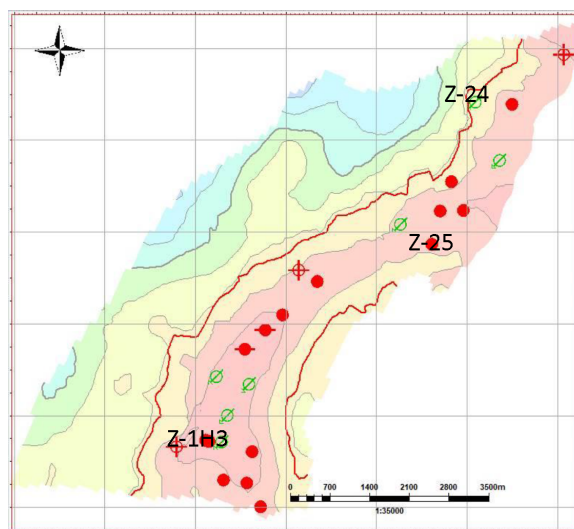


Figure 3—location map of the studied field highlighting three wells of interest (Z-1H3, Z- 24, and Z-25). Red circles indicate producers, and green circles indicate injectors.

The formation pressure data acquired in subsequent wells showed depletion. Thus, OWC is based on the resistivity and saturation profile only. The deepest Z A2 penetration in the field is the well Z-1H3. The Z A2 initial base case OWC is X776 m TVDSS interpreted at the base of high porous interval where resistivity and hydrocarbon saturation reduced towards the bottom. However, based on the Picket plot (Fig.5) relatively tight reservoir could be oil bearing down to X825 m TVDSS. In addition, gas shows from mud log below X776 m TVDSS support deep contact possibility as well (Fig.4).

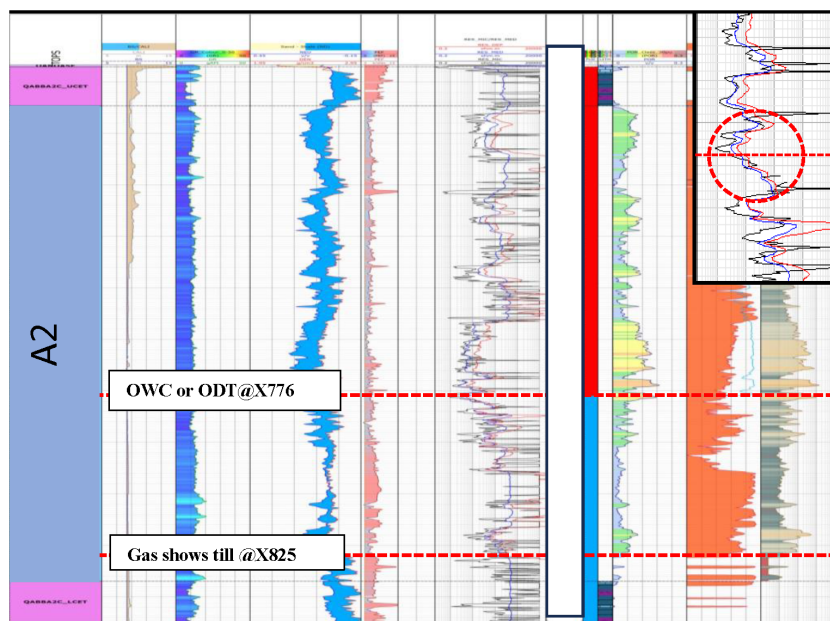


Figure 4—Z-1H3 OWC at X776 m TVDSS and deeper, ODT at X825 m TVDSS, track 1: Tops, track 2: BS & CALI, track 3: GR, track 4: DEN & NEU, Track 5: PEF, Track 7: RES, Track 9: Fluid Flag, Track 11: Por, Track 12 Saturation, Track 13, Permeability, Zoom out of resistivity log in dashed red circle showed slight drop in resistivity log in a good porosity interval.

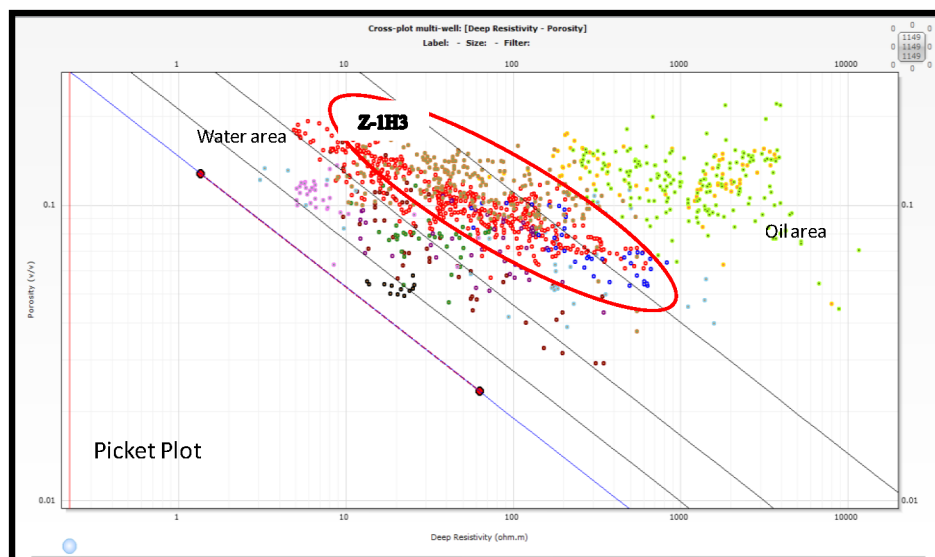


Figure 5—Picket Plot Z-1H3: red dots - points above X776 m TVDSS, Blue points below X776 m TVDSS. Green and Brown points are from oil bearing part in different wells

In 2024, well Z-24H1 was drilled to a deeper structural position with the objective of appraising the uncertainty surrounding the Oil- Water Contact (OWC) (Fig.3). This well indicated a deeper Oil Down To (ODT) at X810 mTVDSS, Subsequently, in 2025, well Z- 25H1 was drilled and it established the OWC at X766 mTVDSS—44 meters shallower than the ODT observed in Z-24H1.

To investigate the observed variation in oil-water contact (OWC) across the Z field, a systematic, multi-disciplinary approach was employed. The workflow integrates resistivity log interpretation, geological and petrophysical analysis, and scenario-based reservoir modelling. The key components of the methodology are outlined & explained below.

OWC Interpretation from Resistivity Logs

The primary tool for OWC interpretation was the resistivity log response across multiple appraisal and development wells. The transition from high to low resistivity was used to delineate the oil-bearing interval and infer fluid contact depths.

Z-24H1 was drilled in 2024 with the objective of appraising the initial oil-water contact (OWC). The well indicated an Oil Down To (ODT) at X810 mTVDSS (Fig. 6). ODT refers to the deepest depth at which movable oil is observed in the reservoir before the reservoir quality significantly deteriorates. In the case of Z-24H1, the reservoir transitions into a low-porosity, tight interval below this depth, which prevents reliable identification of a definitive oil-water contact. As a result, only an ODT could be determined, rather than a clear OWC Z-25H1, drilled a few months after Z-24H1, revealed a clear Oil-Water Contact (OWC) at X766 mTVDSS — which is 44 meters shallower than the Oil Down To (ODT) observed in Z-24H1. The OWC represents the depth at which the reservoir transitions from oilbearing to water-bearing. In Z-25H1, this contact is clearly identified by a distinct drop in resistivity on the log, coinciding with good reservoir properties — including adequate porosity and favorable mobility. These indicators collectively confirm the presence of mobile water below X766 mTVDSS, establishing a definitive OWC in this well (Fig. 7)

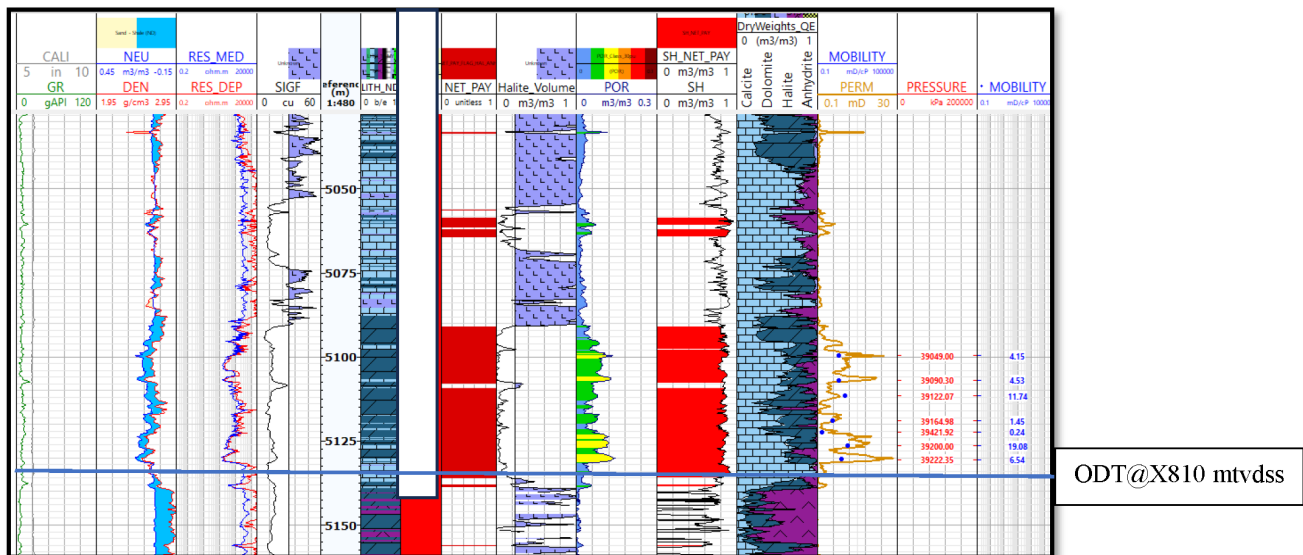


Figure 6—Z-24H1 showed deeper ODT at X810 mTVDSS compared to initial OWC defined in Z-1H3; track 1: GR & Cali, track2: DEN & NEU, Track 3: RES MED & RES DEP, Track 4: Formation Sigma (used mainly halite plugged intervals), track 6: Lithology track from DEN & NEU, track 7: Fluid Flag, Track 8: Net Pay Flag, Track 9: Halite volume, Track 10: POR, Track 11: HC Saturation, Track 12: Lithology Track, Track 13: Perm & Mobility Track 14 & 15: Pressure & mobility

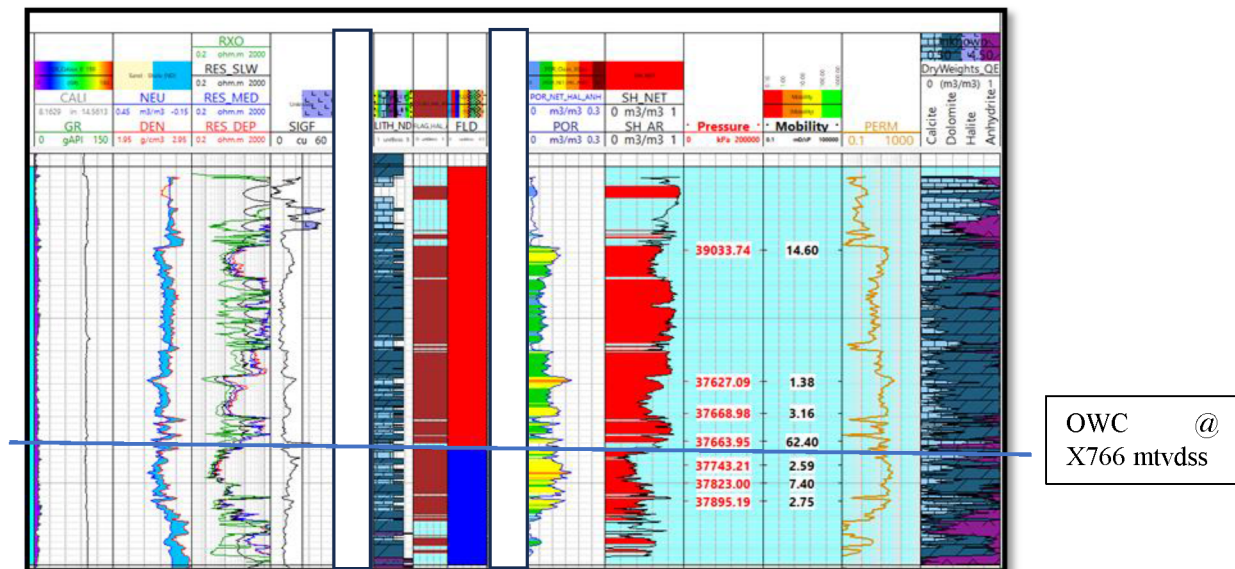


Figure 7—Z-25H1 showed clear OWC 44 m shallower than the ODT seen in Z-24H1; track 1: GR & Cali, track2: DEN & NEU, Track 3: Resistivity, Track 4: Formation Sigma (used mainly halite plugged intervals), track 6: Lithology track from DEN & NEU, track 7: Net Pay Flag, Track 8: Fluid Flag (red indicate oil and blue indicate water), Track 9: POR, Track 10: HC Saturation, Track 11, 12 & 13: pressure, mobility & permeability respectively

Each well was individually analyzed to determine the resistivity cutoff values consistent with hydrocarbon presence, while accounting for variations in porosity, and potential mud filtrate invasion. Cross-plots of resistivity versus TVDSS colored by porosity, were used to validate the contact interpretation. (Fig. 8). To minimize the impact of formation tightness on resistivity measurements, porosity values were filtered so that only data points with porosity greater than 10 porosity unit (p.u.) were included in the plot.

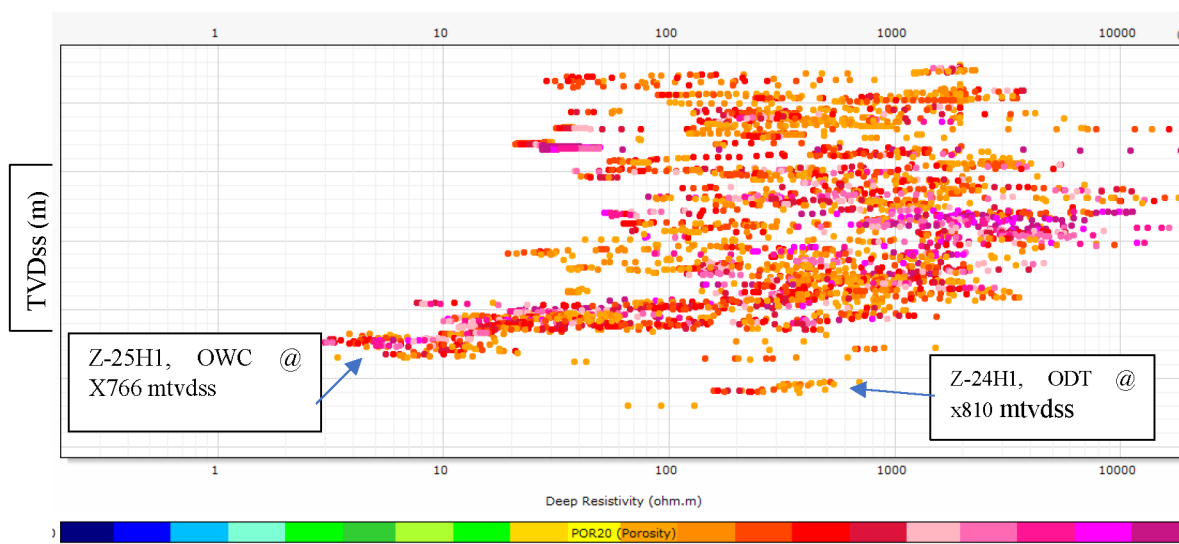


Figure 8—TVDSS vs. Resistivity Plot (filtered with porosity > 10 p.u)

Incorporating both pressure data and production logging data

While petrophysical logs provide an initial estimate of the contact, integrating dynamic data significantly enhances interpretation accuracy. Pressure data, particularly from Wireline Formation Tests, are critical for identifying fluid gradients and detecting changes in fluid phases.

Formation pressure data acquired from wells in the Z Field have been instrumental in confirming a consistent oil gradient across the reservoir. However, establishing a reliable water gradient has been more challenging. Only one well penetrated the water leg before reservoir depletion, but the pressure data acquired from this zone were compromised due to formation tightness. The low permeability of the rock led to supercharging during the pressure test, resulting in artificially elevated readings that do not reflect true formation pressure. As a result, the data from this interval were inconclusive and could not be confidently used to define the original water gradient. The Z-25H1 well also penetrated the water leg, where pressure data was acquired. The measured gradient is approximately 11 kPa/m, confirming the water gradient.

Moreover, Production Logging Tools (PLTs) are used to evaluate fluid movement within the wellbore and the reservoir during production or injection. These logs help determine which zones are contributing fluids and what type of fluids (oil, gas, or water) are being produced. The Production Logging Tool (PLT) data for Z-24H1 was reviewed to validate the Oil Down To (ODT) depth identified from the open-hole logs. The PLT results confirmed the presence of mobile oil down to X810 mTVDSS, supporting the interpretation of the ODT in this well (Fig. 9)

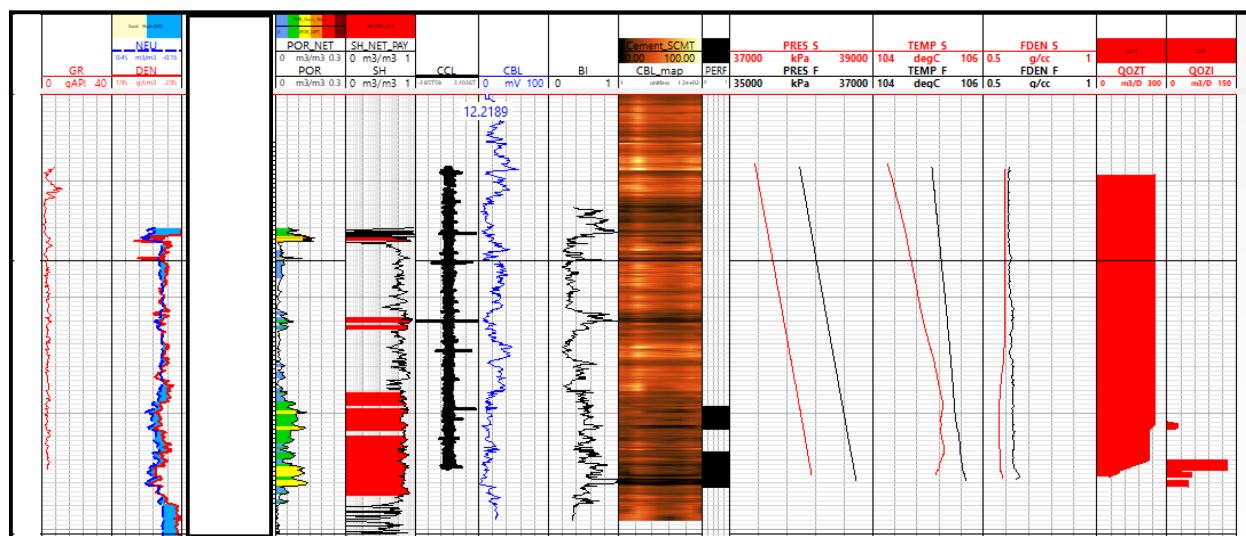


Figure 9—Production logging results for Well Z-24, showing pressure, temperature, flow, and density responses across perforated intervals, with indications of fluid entry and zonal contribution, last track showed 100 % oil production up down to X810mtdvss

Additionally, the PLT data for Z-25H1 was reviewed. During the shut-in pass, a stagnant water level was observed just below the bottom perforation, while the flowing pass indicated active water movement across the bottom perforation and relatively stable flow across the top perforation. It is important to note that the spinner sensor was not operational during the run; therefore, the interpretation was based on capacitance and density sensor readings, as well as temperature anomalies. Temperature logs (Fig. 10) show heating effects across both perforations, which may indicate minor water production. Furthermore, a drop in resistivity at the top perforation—observed in the open-hole logs—suggests possible hydrocarbon desaturation, supporting the interpretation of water production at that interval.

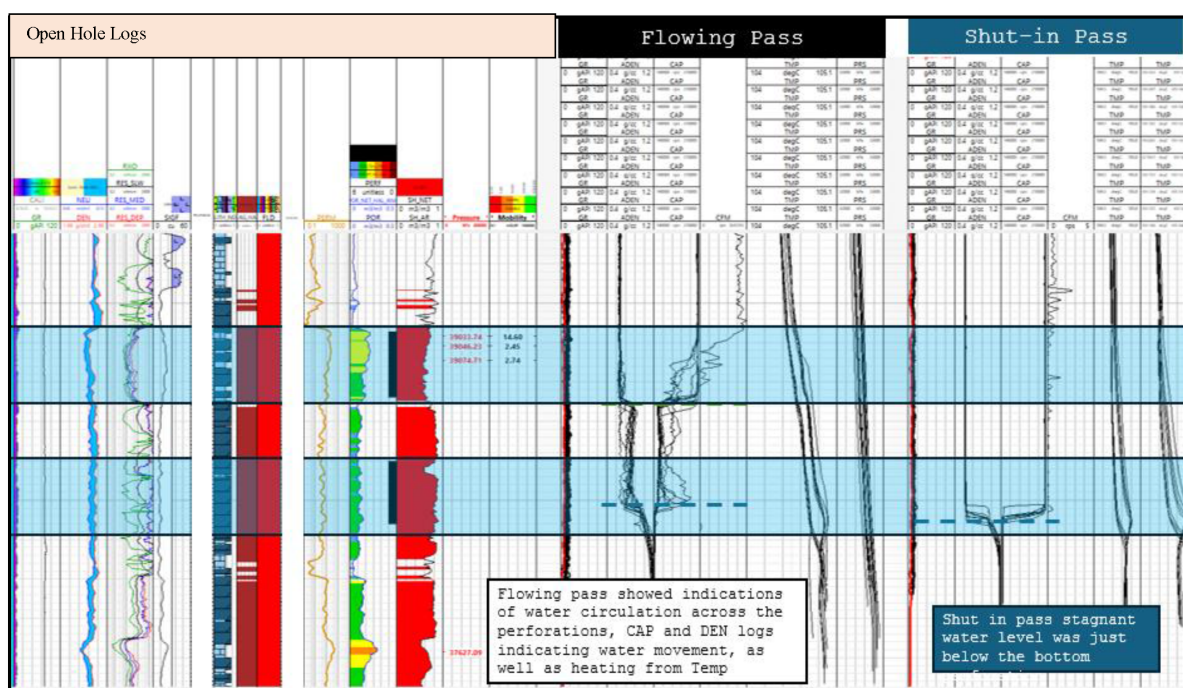


Figure 10—MPLT data from Z-25H1 showing active water circulation during flowing and stagnant water level just below the bottom perforation during shut-in.

Production data further supports these observations by indicating the nature of produced fluids over time. All wells —except Z-25H1— exhibited sustained dry oil production. In contrast, Z-25H1 showed early water breakthrough during well testing, which aligns with the indications from both the open-hole and PLT logs. (Fig.11)

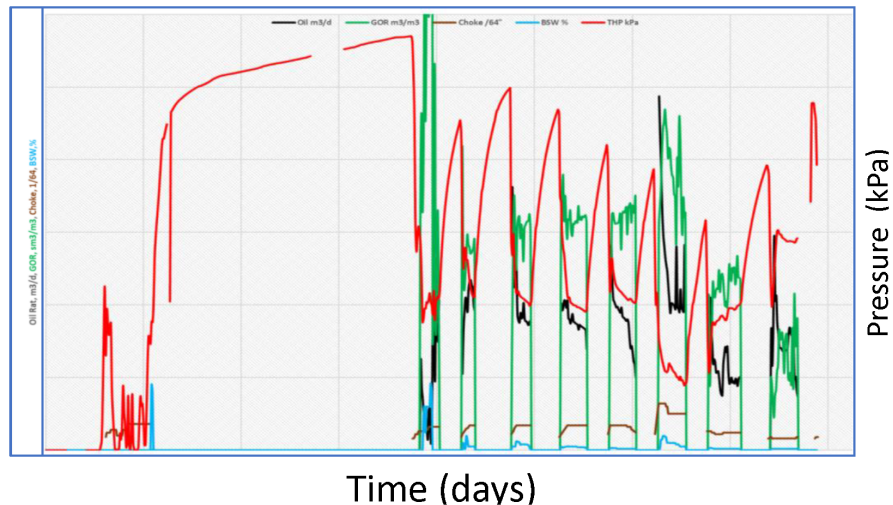


Figure 11—Production test of Well Z-25 showing variations in oil rate, gas-oil ratio (GOR), choke size, bottom hole solids and water (BS&W), and tubing head pressure (THP) over time. It indicates some water production during well testing.

Geological and Structural Framework Construction

An updated structural model of the Z field was built using interpreted seismic data, well tops, and fault picks. The structural framework included the main north-south trending anticline, associated strike-parallel faults, and observed east-west cross-faults, particularly at the field's extremities. Fault geometry and offset along with pressure data and production data were incorporated to assess the possibility of compartmentalization scenario in the field. Several faults have also been interpreted across the field however these faults are short with low throw and unlikely to cause compartmentalization in the field. Special attention was given to the location of Z-25H1 in relation to Z-24H1 to understand the structural context of the observed fluid contact anomaly. Given that the structure of the field is an asymmetrical anticline, the possibility of different OWC on the east and west sides of the structure with possible perched OWC on the east side has been hypothesized. On the southeast part of the structure the A2 may be directly in contact with the A1 stringer (if this is the case it would mean perched water on the east side will not be the case).

Perched Water Concept

Perched water in petroleum reservoirs is defined as mobile formation water that became trapped locally during hydrocarbon migration and is disconnected from the regional aquifer. This typically occurs when structural or stratigraphic barriers prevent complete drainage of connate water during charge, resulting in a secondary, shallower oil-water contact (OWC) above the field-wide equilibrium contact (Macaluso et al., 2019). Early work by Johnson et al. (1986) in the Fulmar field and subsequent studies highlighted that perched accumulations are small in volumetric footprint and sustained only by weak aquifer drive, yet they can persist for long production times (Gaafar et al., 2016; Hulea, 2018). Modern terminology distinguishes between "perched water" and the broader concept of "excess water," the latter encompassing perched aquifers, associated transition zones, and remnant transition intervals left behind after partial drainage (Larsen et al., 2020; Colin et al., 2019). Excess water can be static (no further flow to the aquifer) or dynamic (slow drainage over geologic time), reflecting capillary-gravity equilibrium states (Rolfsvag & Danielsen, 2016; Larsen et al., 2020).

Mechanically, perched water is characterized by pressure anomalies where multiple water gradients intersect a single hydrocarbon gradient, producing over pressured local aquifers ("perch pressure") relative to the regional water table (Larsen et al., 2020). Capillary pressure and rock quality are controlling factors: high-permeability reservoirs with short transition zones often retain perched contacts because water becomes immobile just above the barrier, whereas poorer-quality rocks with longer transition zones tend to drain more completely (Hulea, 2018; Colin et al., 2019). Perched pools may be confirmed through log-derived saturation-height anomalies, capillary pressure curves, and water sample chemistry, and in some cases direct core recovery of moveable water (Colin et al., 2019). Case studies in both clastic and carbonate settings including the Fulmar, Grane, and Red Fields. Fields in Deepwater West Africa demonstrate that perched intervals are encountered across a variety of depositional systems and are often only identified during drilling and production (Macaluso et al., 2019; Colin et al., 2019).

From a development perspective, perched water complicates fluid contact interpretation and volumetric assessments. Because perched zones can mimic field-wide OWCs if misidentified, they risk underestimating hydrocarbons-in-place or misplacing completions (Colin et al., 2019). Pressure and log data showing multiple fluid gradients, combined with tracer production analysis, are key in differentiating perched aquifers from main aquifer drive (Macaluso et al., 2019). Importantly, water production from perched intervals is usually modest and often declines with time as the isolated pool is depleted, meaning their impact on long-term hydrocarbon recovery may be limited but their recognition is vital for accurate modelling (Macaluso et al., 2019; Colin et al., 2019).

As shown in several studies (Yuster, 1953; Hulea, 2018; Dennis, Bergmo, & Holt, 2005), the perched water phenomenon has been recognized in various geological settings, including impermeable barriers, down-dip structural lows, regional facies transitions, and sealing faults (Figs. 12 and 13). Moreover, perched water has been observed in both carbonate and clastic reservoirs.

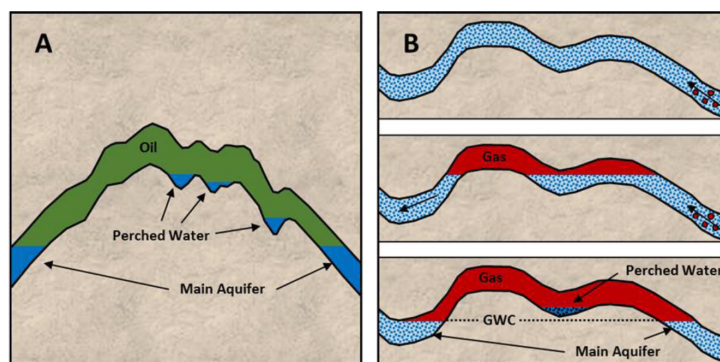


Figure 12—Perched water environments: A & B downcast structures in the bottom of the reservoir (Gamal, et. al.,2016)

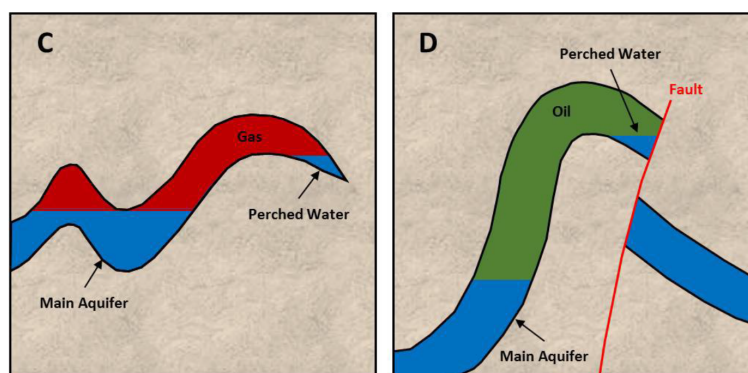


Figure 13—Perched water environments: regional changes in the facies; D Sealing fault (Gamal, et. al.,2016)

Discussion

The significant variation in Oil-Water Contact (OWC) depth observed between wells Z-24H1 and Z-25H1 poses a complex reservoir characterization challenge. Figure 14 presents the stick plot for all wells in the Z Field, illustrating variations in fluid contacts across the structure. This highlights the complexity of the reservoir architecture, possibly influenced by faulting or stratigraphic variations leading to multiple oil-water contacts within the field. The presence of these varying contact depths may indicate perched oil-water contacts or separate fluid compartments, which are important considerations for reservoir management and development planning.

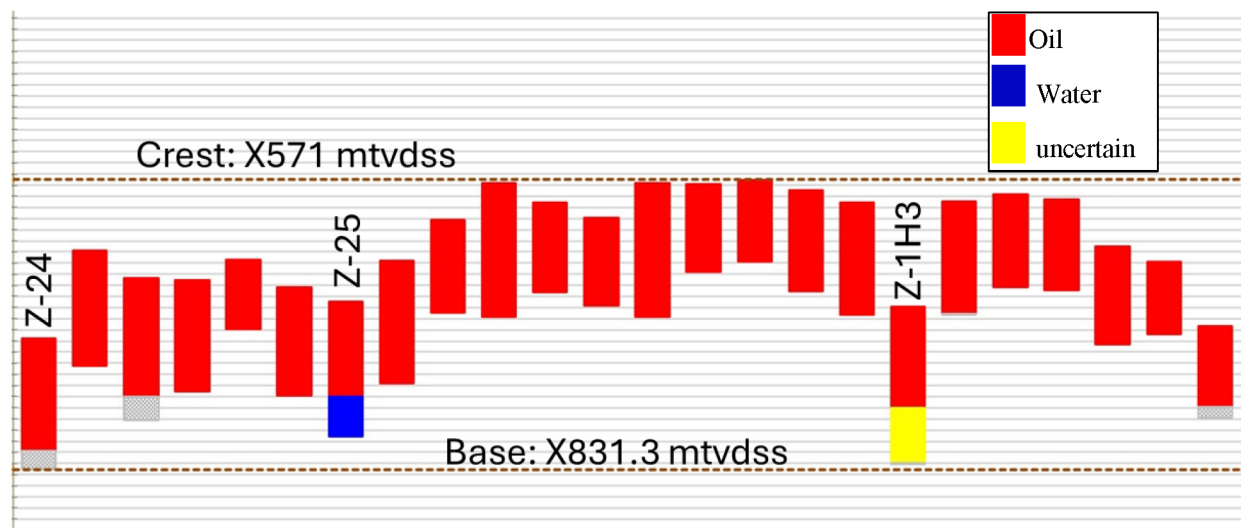


Figure 14—Stick plot showing interpreted fluid contacts across all field wells, plotted against true vertical depth subsea (TVDSS). The structural top reservoir is at X571 mTVDSS and the base at X831.3 mTVDSS. Emphasis is placed on wells Z-24, Z-25, and Z-1H3, which display distinct fluid contact indicators highlighted in color.

To rationalize this discrepancy, three possible scenarios have been evaluated, supported by petrophysical and pressure data analyses: These scenarios are summarized in the table below:

Table 1—Different Possible Scenarios to explain the OWC contact variation

Scenario	BC_OWC (mTVDss)	Evidence	Probability
Scenario 1 (one block and Z-25 is impacted by desaturation)	X810	Good reservoir properties in Z-25 with high mobility / and high production from the northern area	Very unlikely
Scenario 2 (Perched water impacting Z-25 area)	X810	Perched contact is common in pinch out scenario, in Z the eastern flank is fully isolated by salt, and it has shorter limb compared to the western part, therefore it limits the water displacement in the eastern flank	Possible
Scenario 3 (Z-24 area is isolated block) / pressure data does not support this concept	X766	Pressure data from well Z-24 indicates reservoir depletion, which does not support the hypothesis of a separate block	very unlikely

Scenario 1: Local Desaturation Effects Due to Nearby Production

One hypothesis considers the impact of local reservoir pressure depletion and fluid phase redistribution resulting from high production rates in adjacent wells. This local dewatering could potentially lead to a transient upward shift in the OWC near the wellbore of Z-25H1. However, quantitative assessment of reservoir pressure and fluid saturation profiles suggests that dewatering alone cannot justify the magnitude

of the observed OWC variation. Dewatering typically causes only minor vertical shifts in fluid contacts, insufficient to explain the 44-meter differential observed. Therefore, this scenario is considered improbable.

Scenario 2: Perched Oil-Water Contact Controlled by Fault-Related Heterogeneity

The Z Field is a highly faulted structure, with significant faulting particularly evident on the southwest flank. Previous interpretations hypothesized the possibility of differing oil-water contacts (OWCs) on the eastern and western flanks of the structure, including the potential for a perched OWC on the eastern side. This hypothesis has been recently supported by data from well Z-25H1, which confirmed the presence of a shallow, perched OWC in the eastern fold limb.

Perched OWC is a common phenomenon in faulted fold structures (Yuster, 1953; Hulea, 2018; Dennis, Bergmo, & Holt, 2005), especially those exhibiting asymmetrical geometry with fold limbs of differing lengths and dips. In such cases, hydrocarbon migration typically displaces formation water on both sides of a fault. However, when a fault acts as a barrier on the shorter fold limb such as on the eastern side of Z Field it can trap water, preventing it from being fully displaced. On the opposite fold limb, hydrocarbons continue to migrate and displace water, resulting in the development of a deeper, continuous OWC. In Z-25H1, the observed shallow contact reflects this perched behavior. Contrary to earlier assumptions that the A2 and A1 stringers may be in direct contact in the southeastern part of the field an interpretation that would preclude perched water. The new data indicates that such connectivity is not present in this area, further supporting the existence of perched water on the eastern flank (Fig. 15).

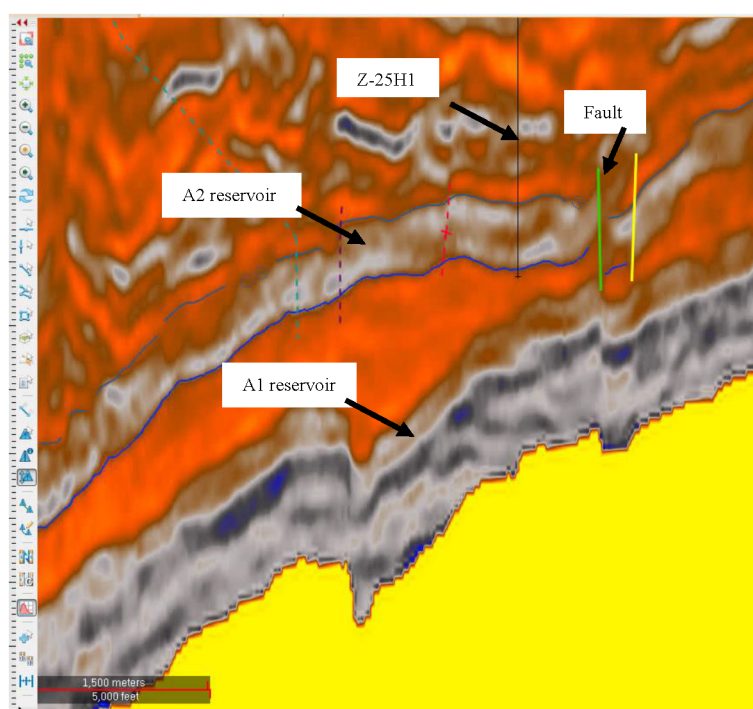


Figure 15—Seismic cross-section data (acoustic impedance) indicates that the A2 and A1 reservoirs are not in communication in the Z-25H1 area. Additionally, the presence of a fault on the opposite side supports the interpretation of a perched water contact in this part of the field

Scenario 3: Reservoir Compartmentalization

The third hypothesis postulates that Z-24H1 and Z-25H1 reside in separate pressure compartments, potentially isolated by fault sealing or stratigraphic barriers, which could explain the differing OWC levels. However, pressure transient analyses and production data from Z-24H1 & Z-25H1 indicate ongoing reservoir depletion, implying pressure communication between the two wells. This evidence strongly disfavors compartmentalization as the cause of the OWC discrepancy

Conclusion and Recommendation

This study provides a novel interpretation of perched water and variable oil-water contacts (OWCs) within the context of intrasalt carbonate stringer reservoirs, using the Z field as a case study. The findings highlight the significant impact of structural compartmentalization, salt isolation, and reservoir geometry on subsurface fluid distribution and support optimized field development strategies under uncertainty.

By integrating resistivity log analysis, structural interpretation, pressure data, and scenario-based reservoir modelling, the study demonstrates that apparent contact anomalies such as shallower OWCs in structurally up dip wells can be explained by perched water or dynamic desaturation effects. In particular, the presence of salt-sealed compartments and fault-bounded fold limbs can lead to localized fluid behaviors that deviate from regional trends, underscoring the importance of considering both geological and dynamic factors in contact interpretation.

Recognizing and correctly identifying perched contacts not only improves the accuracy of oil-in-place estimates, but also enables more informed well placement, depletion planning, and field development strategies. Failure to account for such complexities may lead to underestimation of reserves or suboptimal production decisions.

Overall, this work underscores the value of a multi-disciplinary, integrated approach to reservoir characterization—combining geological, petrophysical, structural, and dynamic data to reduce subsurface uncertainty. These insights are particularly relevant for complex carbonate-salt systems, where isolation and pressure discontinuities are common, and conventional interpretations may fall short.

Acknowledgement

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Nomenclature

OWC	Oil- Water Contact
ODT	Oil Down To
PLT	Production Logging Tool
kPa	Kilopascal
MGI	Miscible Gas Injection
TST	Transgressive Systems Tracts
HST	High stand Systems Tracts
P.u	Porosity Unit
POR	Porosity
RES	Resistivity
DEN & ENU	Density & Neutron
PERM	Permeability

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